Year 2 Pure Mathematics

Chapter 12

Vectors

Practice Booklet

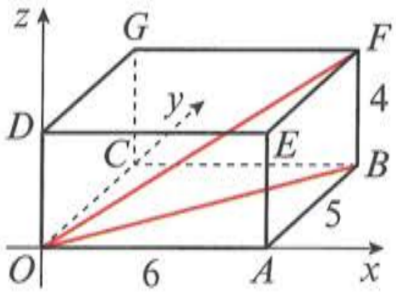
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12.1 3D coordinates

Practice Book, printed page 105.

1. The rectangular prism $OABCDEFG$ is shown in the diagram, positioned so that O lies at the point $(0, 0, 0)$.



a. Use Pythagoras' theorem to calculate the exact length of:

i. OB

ii. OF

The coordinates of point F are (p, q, r).

b. Write down the values of p, q and r.

c. Calculate $\sqrt{p^2 + q^2 + r^2}$.

d. Compare your answers to parts **a ii** and **c**.

2. Calculate, leaving your answer as a simplified surd, the distance from the origin to the point:

a. $A(4, 10, 3)$

b. $B(6, 4, -8)$

c. $C(-2, -4, 15)$

Hint

The distance between the origin and the point (x, y, z) is $\sqrt{x^2 + y^2 + z^2}$

3. In each case, calculate the distance between the points P and Q:

a. $P(5, 3, -2)$ and $Q(-4, 5, 4)$

b. $P(-1, -3, 6)$ and $Q(9, -14, 4)$

c. $P(11, -2, -1)$ and $Q(7, 0, 4)$.

Hint

The distance between points (x_1, y_1, z_1) and (x_2, y_2, z_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

4. The coordinates of A and B are $(6, -1, 3)$ and $(3, 5, k)$ respectively.
Given that the distance from A to B is 7 units,

a. show that $(3 - k)^2 = 4$.

b. Hence find the possible values of k.

Hint

Substitute your coordinates for A and B into the formula for the distance between two points.

5. E The coordinates of A and B are $(-2, 4, 8)$ and $(0, k, 7)$ respectively.
Given that the distance from A to B is 3 units, find the possible values of k. (3 marks)

6. E The coordinates of A and B are $(2, -5, k)$ and $(-2, -1, 3)$ respectively.
Given that the distance from A to B is 9 units, find the possible values of k. (3 marks)

7. E The coordinates of A and B are $(k, -4, 6)$ and $(8, -5, -1)$ respectively.
Given that the distance from A to B is $5\sqrt{3}$ units, find the possible values of k. (3 marks)

8. E The coordinates of A and B are $(4, -4, 3)$ and $(k, 4, 4)$ respectively.
Given that the distance from A to B is $3\sqrt{10}$ units, find the possible values of k. (3 marks)

12.2 Vectors in 3D

Practice Book, printed page 106.

1. Consider the points $A(4, -3, 1)$ and $B(-2, 6, 5)$.

a. Write the position vector of A in the form $p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$.

b. Write the position vector of B in the form $\begin{pmatrix} p \\ q \\ r \end{pmatrix}$.

Hint

$\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors along the x-, y- and z-axes.

$$p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

2. P is the point $(5, 2, 1)$ and Q is the point $(3, 7, -2)$.

a. Find the vector $\overrightarrow{PQ}$ giving your answer:

i. in **ijk** notation

ii. as a column vector.

b. Find the distance between P and Q.

c. Find the unit vector in the direction of $\overrightarrow{PQ}$.

Hint

For the vector $\mathbf{a}$, where $\mathbf{a} = p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$, the unit vector, $\hat{\mathbf{a}}$, is given by $\hat{\mathbf{a}} = \dfrac{p\mathbf{i} + q\mathbf{j} + r\mathbf{k}}{\sqrt{p^2 + q^2 + r^2}}$

3. The vectors $\mathbf{m}$ and $\mathbf{n}$ are defined by $\mathbf{m} = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{n} = \begin{pmatrix} -4 \\ -5 \\ 6 \end{pmatrix}$.

a. Find, giving your answer in the form $p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$:

i. $\mathbf{m} + 2\mathbf{n}$

ii. $-\mathbf{m}-\mathbf{n}$

b. Show that the vector $4\mathbf{m}-\mathbf{n}$ is parallel to the vector $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$.

Hint

Two vectors are parallel if one is a scalar multiple of the other.

4. Find, to one decimal place, the angle that the vector $\mathbf{a}=2\mathbf{i}-\mathbf{j}-2\mathbf{k}$ makes with:

a. the positive x-axis

b. the positive y-axis

c. the positive z-axis.

Hint

If the vector $\mathbf{a}=x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$ makes an angle θ_x with the positive x-axis then $\cos\theta_x=\dfrac{x}{|\mathbf{a}|}$ and similarly for the angles θ_y and θ_z.

5. P Given that $\overrightarrow{OA}=2\mathbf{i}+3\mathbf{j}-5\mathbf{k}$ and $\overrightarrow{OB}=-\mathbf{i}+4\mathbf{j}+2\mathbf{k}$,

a. find $\overrightarrow{AB}$

b. show that the triangle OAB is scalene.

6. E Given $\overrightarrow{AB}=4\mathbf{i}+\mathbf{j}+3\mathbf{k}$ and $\overrightarrow{AC}=5\mathbf{i}-6\mathbf{j}-\mathbf{k}$, find the unit vector in the direction $\overrightarrow{BC}$.(3 marks)

7. E/P The points A, B and C have position vectors $\begin{pmatrix}4\\4\\-3\end{pmatrix}$, $\begin{pmatrix}7\\-1\\0\end{pmatrix}$ and $\begin{pmatrix}-1\\1\\-6\end{pmatrix}$ respectively.

a. Find the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BC}$. (3 marks)

b. Find $|\overrightarrow{AB}|$, $|\overrightarrow{AC}|$ and $|\overrightarrow{BC}|$ giving your answers in exact form. (3 marks)

c. Describe the triangle ABC. (1 mark)

8. E/P Triangle PQR has vertices $P(1,2,-3)$, $Q(1,2,3)$ and $R(-2,-2,-3)$.

a. Find $|\overrightarrow{PQ}|$, $|\overrightarrow{QR}|$ and $|\overrightarrow{PR}|$ giving your answers in exact form. (3 marks)

b. Use your answer to part **a** to explain why $\angle RPQ$ is a right angle. (2 marks)

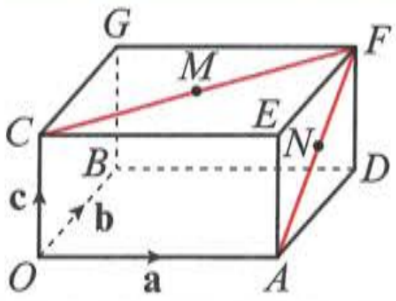
c. Find the size of angle PRQ, giving your answer to one decimal place. (2 marks)

9. E Given $\overrightarrow{AB}=-\mathbf{i}+2\mathbf{j}+\mathbf{k}$ and $\overrightarrow{AC}=-8\mathbf{i}+7\mathbf{j}+2\mathbf{k}$ in triangle ABC, find $\angle ABC$, giving your answer in degrees to one decimal place. (6 marks)

12.3 Solving geometric problems

Practice Book, printed page 107.

1. The diagram shows a rectangular prism.
The vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are equal to **a**, **b** and **c** respectively. M and N are the midpoints of CF and AF respectively. Express in terms of **a**, **b** and **c**:

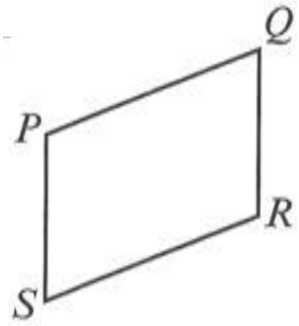


a. $\overrightarrow{OF}$
b. $\overrightarrow{DC}$
c. $\overrightarrow{AG}$
d. $\overrightarrow{CF}$
e. $\overrightarrow{OM}$
f. $\overrightarrow{AM}$
g. $\overrightarrow{AN}$
h. $\overrightarrow{NM}$

Hint

Use the rules for vector addition, working in three dimensions.

2. The diagram shows a quadrilateral $PQRS$, where P, Q, R and S are the points $(7, 6, 3)$, $(4, 10, -4)$, $(3, 15, -12)$ and $(6, 11, -5)$ respectively.



Find:
a. $\overrightarrow{PQ}$ and $\overrightarrow{SR}$
b. $\overrightarrow{SP}$ and $\overrightarrow{RQ}$
c. Describe the quadrilateral $PQRS$.

3. Given that

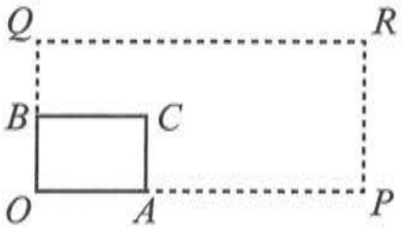
$$(2q - 3p)\mathbf{i} + 5\mathbf{j} + r\mathbf{k} = 13\mathbf{i} + (4q + p)\mathbf{j} + (2q - p)\mathbf{k}$$

find the values of p, q and r.

Hint

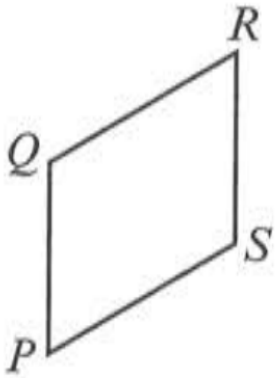
As the vectors **i**, **j** and **k** are non-coplanar, you can compare coefficients.
If $a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = d\mathbf{i} + e\mathbf{j} + f\mathbf{k}$, then $a = d$, $b = e$ and $c = f$.

4. A rectangle $OACB$ has an area of 20. A new rectangle $OPRQ$ is formed, such that $\overrightarrow{OP} = 3\overrightarrow{OA}$ and $\overrightarrow{OQ} = 2\overrightarrow{OB}$.
Find the area of the new rectangle.



5. E Given that $-a\mathbf{i} - 6\mathbf{j} + (3a + 2b)\mathbf{k} = 2b\mathbf{i} + bc\mathbf{j} + 8\mathbf{k}$, find the values of a, b and c. (4 marks)

6. E The diagram shows the parallelogram $PQRS$.



Given that $\overrightarrow{PQ} = \mathbf{i} + 10\mathbf{j} - 3\mathbf{k}$ and $\overrightarrow{QR} = 8\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$, find the area of the quadrilateral, giving your answer to one decimal place. (6 marks)

7. P The vertices of triangle ABC have coordinates $A(2, 11, 4)$, $B(6, k, -6)$ and $C(12, 6, 1)$.
Given that triangle ABC is isosceles, and k is a positive integer, find the value of k.

8. E/P A student makes the following hypothesis:

> It is not possible to have an equilateral triangle in 3 dimensions with each vertex having integer coordinates.

Show that the student's hypothesis is incorrect. (5 marks)

12.4 Applications to mechanics

Practice Book, printed page 108.

1. A particle is acted on by three forces $\mathbf{F}_1 = (-5\mathbf{i} + 3\mathbf{j} + \mathbf{k})$ N, $\mathbf{F}_2 = (2\mathbf{i} - 5\mathbf{k})$ N and $\mathbf{F}_3 = (-4\mathbf{j} + 3\mathbf{k})$ N.

a. Find the resultant force $\mathbf{R}$.

b. Find the magnitude of the resultant force.

Hint

The resultant force can be found by summing all three forces.
The magnitude of the force $\mathbf{F} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is $\sqrt{a^2 + b^2 + c^2}$.

2. A particle is in equilibrium and is acted on by three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where $\mathbf{F}_1 = (2b\mathbf{i} - b\mathbf{j} + 5\mathbf{k})$ N, $\mathbf{F}_2 = (-4\mathbf{i} + 7\mathbf{j} - 8\mathbf{k})$ N and $\mathbf{F}_3 = (a\mathbf{i} + 2a\mathbf{j} + 3\mathbf{k})$ N.
Find the values of a and b.

Hint

If the particle is in equilibrium, the resultant force will be zero.

3. A particle, initially at rest at O, is acted on by a force that causes the particle to accelerate at $(-3\mathbf{i} + 6\mathbf{j} - \mathbf{k})\,\mathrm{m\,s^{-2}}$ for 2 seconds.

a. Find the magnitude of the acceleration of the particle.

b. Find the distance of the particle from O after 2 seconds.

Hint

Use the suvat equations to find the position of the particle.
Here, $s = ut + \frac{1}{2}at^2$ is the correct formula to use.
← **Statistics and Mechanics Year 1, Chapter 9**

4. A particle of mass 4 kg is acted on by two forces, $\mathbf{F}_1 = (8\mathbf{i} - 3\mathbf{j} + 6\mathbf{k})\,\mathrm{N}$ and $\mathbf{F}_2 = (-6\mathbf{i} + 9\mathbf{j} + 4\mathbf{k})\,\mathrm{N}$.

a. Find the resultant force acting on the particle.

b. Find the acceleration of the particle, giving your answer as a vector.

Hint

Use the vector form of Newton's second law: $\mathbf{F} = m\mathbf{a}$.
← **Statistics and Mechanics Year 1, Section 10.4**

5. E A particle, initially at rest at O, accelerates at $(5\mathbf{i} - 11\mathbf{j} + 2\mathbf{k})\,\mathrm{m\,s^{-2}}$ for 4 seconds.
Find:

a. the speed of the particle after 4 seconds, giving your answer in the form $a\sqrt{6}\,\mathrm{m\,s^{-1}}$, where a is a constant (3 marks)

b. the distance travelled by the particle in the first 4 seconds of its motion. (2 marks)

6. E Given that a force of $(5\mathbf{i} - 6\mathbf{j} + 12\mathbf{k})\,\mathrm{N}$ acts on a particle of mass 4 kg, find:

a. the acceleration of the particle (2 marks)

b. the angle the acceleration vector makes with $\mathbf{i}$. (3 marks)

7. E/P A particle of mass 3 kg is in static equilibrium and is acted on by three forces $\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j} + b\mathbf{k})\,\mathrm{N}$, $\mathbf{F}_2 = (a\mathbf{j} + 4\mathbf{k})\,\mathrm{N}$ and $\mathbf{F}_3 = (-2\mathbf{i} - 5\mathbf{j} + 3\mathbf{k})\,\mathrm{N}$.

a. Find the values of the constants a and b. (2 marks)

The force $\mathbf{F}_3$ is removed. Find:

b. the resultant force $\mathbf{R}$ (1 mark)

c. the acceleration of the particle, giving your answer in the form $(p\mathbf{i} + q\mathbf{j} + r\mathbf{k})\,\mathrm{m\,s^{-2}}$ (2 marks)

d. the magnitude of the acceleration. (2 marks)

8. E/P A particle of mass 5 kg is acted on by three forces, $\mathbf{F}_1 = (2\mathbf{i} + p\mathbf{j} - \mathbf{k})$ N, $\mathbf{F}_2 = (p\mathbf{i} + 5\mathbf{j} + 2\mathbf{k})$ N and $\mathbf{F}_3 = (\mathbf{i} + \mathbf{j} - p\mathbf{k})$ N. Given that the particle is accelerating and that the magnitude of that acceleration is $5\,\mathrm{m\,s^{-2}}$, find the possible values of p. (5 marks)

Problem solving: Set A

Bronze

The coordinates of A and B are $(8, -6, 4)$ and $(k, -4, -7)$ respectively.
Given that the distance from A to B is 15 units, find the possible values of k. (3 marks)

Silver

Given that $\mathbf{a} = 4p\mathbf{i} - 5p\mathbf{j} + 3p\mathbf{k}$, and that $|\mathbf{a}| = 10\sqrt{6}$, find the possible values of p. (3 marks)

Gold

A particle of mass 3 kg is acted on by three forces:
$\mathbf{F}_1 = (4\mathbf{i} + q\mathbf{j} + 2\mathbf{k})$ N, $\mathbf{F}_2 = (-q\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ N and $\mathbf{F}_3 = (-\mathbf{i} + 3\mathbf{j} - q\mathbf{k})$ N.
Given that the particle is accelerating and that the magnitude of that acceleration is $2\,\text{m}\,\text{s}^{-2}$, find the possible values of q. (5 marks)

Problem solving: Set B

Bronze

A, B, C and D are the points $(7, 4, -1)$, $(11, -2, 3)$, $(6, -15, 16)$ and $(-2, -3, 8)$ respectively.

a. Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$, giving your answers in the form $p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$. (2 marks)

b. Prove that $\overrightarrow{AB}$ and $\overrightarrow{DC}$ are parallel. (2 marks)

c. Find k such that $|\overrightarrow{AB}| = k|\overrightarrow{DC}|$. (1 mark)

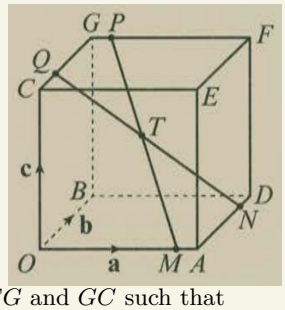
d. Hence describe quadrilateral $ABCD$. (1 mark)

Silver

$ABCD$ is a quadrilateral and A, B, C are the points $(6, -8, 4)$, $(-3, -4, 2)$ and $(-7, 9, -3)$ respectively. Find the coordinates of D such that $ABCD$ is a parallelogram. (4 marks)

Gold

The diagram shows a cuboid whose vertices are O, A, B, C, D, E, F and G.
Vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are equal to $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively.



The points M, N, P and Q lie on OA, AD, FG and GC such that

$$OM : MA = AN : ND = FP : PG = GQ : QC = 4 : 1.$$

Prove that the lines MP and NQ bisect each other at T. (10 marks)

Now try this → Exam question bank Q1, Q13, Q16, Q19, Q65, Q84

Selected exam question bank questions

Question 1

The coordinates of A and B are $(6, 3, -3)$ and $(-4, k, 0)$ respectively.
Given that the distance from A to B is $5\sqrt{5}$ units, find the possible values of k. (3 marks)

Question 13

Given $\overrightarrow{AB} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\overrightarrow{BC} = -7\mathbf{i} - 10\mathbf{j} + 4\mathbf{k}$ in triangle ABC, find $\angle BAC$, giving your answer to 1 decimal place. (6 marks)

Question 16

Given that a force of $(-2\mathbf{i} - 8\mathbf{j} + 4\mathbf{k})$ N acts on a particle of mass 3 kg, find:

a. the acceleration of the particle (2 marks)
b. the angle the acceleration vector makes with the unit vector $\mathbf{j}$. (3 marks)

Question 19

Given that $\mathbf{a} = 2t\mathbf{i} - 4t\mathbf{j} - 2t\mathbf{k}$, and $|\mathbf{a}| = 9\sqrt{6}$, find the possible values of t. (4 marks)

Question 65

In triangle PQR, $\overrightarrow{PQ} = -3\mathbf{i} + 6\mathbf{j} + \mathbf{k}$ and $\overrightarrow{PR} = 11\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$.

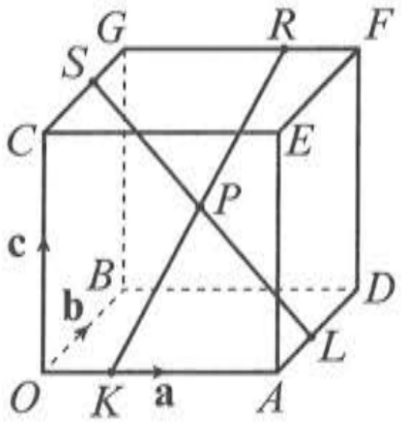
a. Find the area of triangle PQR. (7 marks)

The point T is such that $\overrightarrow{PT} = -2\overrightarrow{PQ}$ and the point S is such that $\overrightarrow{PS} = -2\overrightarrow{PR}$.

b. Find the area of triangle PST. (2 marks)

Question 84

The diagram shows a cuboid whose vertices are O, A, B, C, D, E, F and G.
Vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are equal to $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively.



The points K, L, R and S lie on OA, AD, FG and GC such that

$$OK : KA = AL : LD = FR : RG = GS : SC = 1 : 2.$$

Prove that the lines KR and LS bisect each other at P. (10 marks)

Answers

12.1 3D coordinates

1.

a.

i. $\sqrt{61}$

ii. $\sqrt{77}$

b. $p = 6$, $q = 5$, $r = 4$

c. $\sqrt{77}$

d. They are the same

2.

a. $5\sqrt{5}$

b. $2\sqrt{29}$

c. $7\sqrt{5}$

3.

a. 11

b. 15

c. $3\sqrt{5}$

4.

a. $\sqrt{(6-3)^2 + (-1-5)^2 + (3-k)^2} = 7$
$\Rightarrow 9 + 36 + (3-k)^2 = 49 \Rightarrow (3-k)^2 = 4$

b. $k = 1$, $k = 5$

5. $k = 2$, $k = 6$

6. $k = -4$, $k = 10$

7. $k = 3$, $k = 13$

8. $k = -1$, $k = 9$

12.2 Vectors in 3D

1.

a. $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$

b. $\begin{pmatrix} -2 \\ 6 \\ 5 \end{pmatrix}$

2.

a.

i. $-2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$

ii. $\begin{pmatrix} -2 \\ 5 \\ -3 \end{pmatrix}$

b. $\sqrt{38}$

c. $\frac{1}{\sqrt{38}}(-2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k})$

3.

a.

i. $-6\mathbf{i} - 12\mathbf{j} + 15\mathbf{k}$

ii. $2\mathbf{i} + 7\mathbf{j} - 9\mathbf{k}$

b. $4\mathbf{m} - \mathbf{n} = 4\begin{pmatrix}2\\-2\\3\end{pmatrix} - \begin{pmatrix}-4\\-5\\6\end{pmatrix} = \begin{pmatrix}8+4\\-8+5\\12-6\end{pmatrix} = \begin{pmatrix}12\\-3\\6\end{pmatrix} = 3\begin{pmatrix}4\\-1\\2\end{pmatrix}$

4.

a. $\theta_x = 48.2°$

b. $\theta_y = 109.5°$

c. $\theta_z = 131.8°$

5.

a. $\overrightarrow{AB} = -3\mathbf{i} + \mathbf{j} + 7\mathbf{k}$

b. $|\overrightarrow{OA}| = \sqrt{38}$, $|\overrightarrow{OB}| = \sqrt{21}$ and $|\overrightarrow{AB}| = \sqrt{59}$.
The lengths of the sides are different, so the triangle is scalene.

6. $\dfrac{\mathbf{i} - 7\mathbf{j} - 4\mathbf{k}}{\sqrt{66}}$

7.

a. $\overrightarrow{AB} = \begin{pmatrix}3\\-5\\3\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}-5\\-3\\-3\end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix}-8\\2\\-6\end{pmatrix}$

b. $|\overrightarrow{AB}| = \sqrt{43}$, $|\overrightarrow{AC}| = \sqrt{43}$ and $|\overrightarrow{BC}| = 2\sqrt{26}$.

c. Isosceles

8.

a. $|\overrightarrow{PQ}| = 6$, $|\overrightarrow{QR}| = \sqrt{61}$ and $|\overrightarrow{PR}| = 5$

b. $|\overrightarrow{PR}|^2 + |\overrightarrow{PQ}|^2 = |\overrightarrow{QR}|^2$, so QR is the hypotenuse of a right-angled triangle with $\angle RPQ$ as its right angle.

c. 50.2°

9. 148.1°

12.3 Solving geometric problems

1.

a. $\mathbf{a} + \mathbf{b} + \mathbf{c}$

b. $-\mathbf{a} - \mathbf{b} + \mathbf{c}$

c. $-\mathbf{a} + \mathbf{b} + \mathbf{c}$

d. $\mathbf{a} + \mathbf{b}$

e. $\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} + \mathbf{c}$

f. $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} + \mathbf{c}$

g. $\frac{1}{2}\mathbf{b} + \frac{1}{2}\mathbf{c}$

h. $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}$

2.

a. $\overrightarrow{PQ} = -3\mathbf{i} + 4\mathbf{j} - 7\mathbf{k}$, $\overrightarrow{SR} = -3\mathbf{i} + 4\mathbf{j} - 7\mathbf{k}$

b. $\overrightarrow{SP} = \mathbf{i} - 5\mathbf{j} + 8\mathbf{k}$, $\overrightarrow{RQ} = \mathbf{i} - 5\mathbf{j} + 8\mathbf{k}$

c. $PQRS$ is a parallelogram because there are two pairs of opposite sides that are parallel.

3. $p = -3$, $q = 2$, $r = 7$

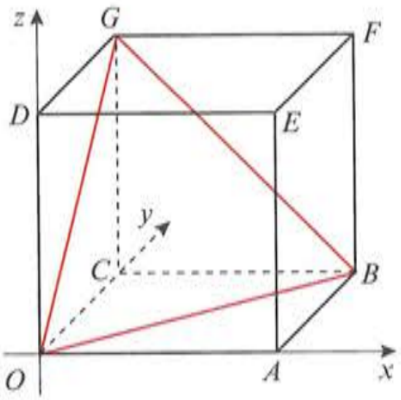
4. 120

5. $a = 4$, $b = -2$, $c = 3$

6. 86.3

7. 13

8. Let $OABCDEFG$ be a cube with side length 5.



$OB = BG = OG = 5\sqrt{2}$, so triangle is equilateral.
$O(0, 0, 0)$, $B(5, 5, 0)$ and $G(0, 5, 5)$, so triangle OBG is equilateral and has integer coordinates.

12.4 Applications to mechanics

1.

a. $\mathbf{R} = (-3\mathbf{i} - \mathbf{j} - \mathbf{k})\,\text{N}$

b. $\sqrt{11}\,\text{N}$

2. $a = -2$, $b = 3$

3.

a. $6.78\,\text{m s}^{-2}$ (3 s.f.)

b. $2\sqrt{46}\,\text{m}$

4.

a. $(2\mathbf{i} + 6\mathbf{j} + 10\mathbf{k})\,\text{N}$

b. $(\frac{1}{2}\mathbf{i} + \frac{3}{2}\mathbf{j} + \frac{5}{2}\mathbf{k})\,\text{m s}^{-2}$

5.

a. $20\sqrt{6}\,\text{m s}^{-1}$

b. $40\sqrt{6}\,\text{m}$

6.

a. $(\frac{5}{4}\mathbf{i} - \frac{3}{2}\mathbf{j} + 3\mathbf{k})\,\text{m s}^{-2}$

b. $69.6°$ (3 s.f.)

7.

a. $a = 2$, $b = -7$

b. $(2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k})\,\text{N}$

c. $(\frac{2}{3}\mathbf{i} + \frac{5}{3}\mathbf{j} - \mathbf{k})\,\text{m s}^{-2}$

d. $\frac{1}{3}\sqrt{38}\,\text{m s}^{-2}$

8. $-\frac{7}{3}$ and -3

Problem solving: Set A

Bronze $k = 18$ or $k = -2$

Silver $p = \pm 2\sqrt{3}$

Gold $q = -1$, $q = \frac{1}{3}$

Problem solving: Set B

Bronze

a. $\overrightarrow{AB} = 4\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$ and $\overrightarrow{DC} = 8\mathbf{i} - 12\mathbf{j} + 8\mathbf{k}$

b. $\overrightarrow{DC} = 2\overrightarrow{AB}$

c. $k = \frac{1}{2}$

d. $ABCD$ is a trapezium.

Silver $(2, 5, -1)$

Gold

Let T_1 be a point on MP. $\overrightarrow{OT_1} = \frac{4}{5}\mathbf{a} + \lambda(-\frac{3}{5}\mathbf{a} + \mathbf{b} + \mathbf{c})$ for some $\lambda \in \mathbb{R}$, $0 \leq \lambda \leq 1$.

Let T_2 be a point on NQ. $\overrightarrow{OT_2} = \mathbf{a} + \frac{4}{5}\mathbf{b} + \mu(-\mathbf{a} - \frac{3}{5}\mathbf{b} + \mathbf{c})$ for some $\mu \in \mathbb{R}$, $0 \leq \mu \leq 1$.

If T lies on MP and NQ then $\overrightarrow{OT}$ satisfies both equations.

Equating terms gives: $\frac{4}{5} - \frac{3}{5}\lambda = 1 - \mu$, $\lambda = \frac{4}{5} - \frac{3}{5}\mu$ and $\lambda = \mu$.

Solve to obtain $\lambda = \mu = \frac{1}{2}$.

Conclude that MP and NQ intersect at point T which is the midpoint of each line and therefore MP and NQ bisect each other at T.

Selected exam question bank answers

Question 1 $k = -1$, $k = 7$

Question 13 $127.1°$

Question 16

a. $(-\frac{2}{3}\mathbf{i} - \frac{8}{3}\mathbf{j} + \frac{4}{3}\mathbf{k})\,\mathrm{m\,s^{-2}}$

b. $151°$ (3 s.f.)

Question 19 $t = \pm 4.5$

Question 65

a. 44.6 (3 s.f.)

b. 178.4 (3 s.f.)

Question 84 Let P_1 be a point on KR, $\overrightarrow{OP_1} = \frac{1}{3}\mathbf{a} + \lambda(\frac{1}{3}\mathbf{a} + \mathbf{b} + \mathbf{c})$, for some $\lambda \in \mathbb{R}$, $0 \leq \lambda \leq 1$.

Let P_2 be a point on LS, $\overrightarrow{OP_2} = \mathbf{a} + \frac{1}{3}\mathbf{b} + \mu(-\mathbf{a} + \frac{1}{3}\mathbf{b} + \mathbf{c})$, for some $\mu \in \mathbb{R}$, $0 \leq \mu \leq 1$.

For a point P which lies on KR and LS, $\overrightarrow{OP}$ satisfies both equations.

Solve to obtain $\lambda = \mu = \frac{1}{2}$. Conclude that lines do intersect, as P lies half way along LS and halfway along KR.